IoT for Product Design

From Smart Objects to Ecosystems
From Data to Better Products

Course Guide & Exercise Roadmap

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1. Course Purpose & Vision

The course introduces product design students to physical computing and IoT using **ESP32** and **Arduino Cloud**, driven by two central goals:

1. **From Smart Objects to Ecosystems:**
   Value emerges from the *relationships* between objects, users, and cloud services — not from isolated gadgets.
2. **From Data to Better Products:**
   Sensor data collected in the field informs iterative redesign, maintenance, and user understanding (closing the loop).

- **Target Audience:** Designers with no prior coding experience.
- **Pedagogy:** Step-by-step progression; every technical step ends with a **Design Reflection** and an empirical **Validation**.

Course Agreement & Ground Rules

What we agree on from day one:

- **Use of AI is perfectly fine:** Leverage generative tools as creative pair-programmers and consultants.
- **Physical reality matters:** At the end, I want to see a physical product made of real, interconnected things.
- **No validation is unacceptable and dangerous:** If it isn't tested with real humans and real signals, it doesn't work.
- **Validation needs measures:** Provide concrete numbers, latency, error counts, and observations , **not** generic sentences.

2. Prototyping Strategy: Three Workspaces

To minimize tool friction while respecting cloud quotas on the free tier, students use three complementary workspaces:

Tool	Role	Why Designers Use It
Wokwi	Browser simulator	Sketching behavior and wiring with zero risk and instant feedback
Arduino IDE 2	Desktop environment	Unlimited local uploads, physical testing, Serial Monitor/Plotter
Arduino Cloud	Connected IoT platform	Cloud variables, mobile control, shared dashboards, data storage

> **The Golden Rule:** Try and debug cheaply in **Wokwi** and the **IDE** first; use **Arduino Cloud** only for the connected milestone.

3. The Running Scenario: "The Connected Studio"

All exercises contribute to a shared, living scenario: bringing a **co-working studio** to life.

- Each student group designs and builds **one studio object**:
  - *Desk Task Lamp*, *Focus Totem*, *Air Quality Monitor*, *Presence Badge*, *Plant Pot*, or *Noise Meter*.
- **Progressive Evolution:**
  1. **Phase A:** The object acts completely alone (**Smart Object**).
  2. **Phase B:** The object connects to the network and dashboards (**Connected Object**).
  3. **Phase C:** Objects cooperate directly through shared variables (**Ecosystem**).
  4. **Phase D:** Objects are deployed, gathering real telemetry to justify redesign (**Data-Driven Design**).

4. The 2×2 Interaction Foundation

The foundational exercises are built on a symmetrical **2×2 matrix of physical computing primitives** requiring **zero external libraries**:

	Discrete / Binary (0 or 1)	Continuous / Modulated (0 … Max)
Input (Sensing / Control)	**Push Button** (`digitalRead`, `INPUT_PULLUP`)	**Potentiometer** (`analogRead`, 0–4095)
Output (Actuation / Feedback)	**LED On/Off** (`digitalWrite`, HIGH/LOW)	**LED Dimmer** (`analogWrite`, PWM 0–255)

1:1 Bridge to Arduino Cloud Widgets:

- Physical Button ⟷ Cloud Switch / Push Widget
- Physical LED ⟷ Cloud Status Light Widget
- Physical Potentiometer ⟷ Cloud Gauge / Percentage Widget
- Physical PWM Dimmer ⟷ Cloud Slider / Dimmer Widget

5. Hardware Kit: Core vs. Project Modules

To prevent beginner frustration (I2C address conflicts, power brownouts, library mismatches), the kit is divided into two tiers:

5.1 Core Kit (Universal — Ex 0 to 2)

- ESP32 DevKit, breadboard, jumper wires, USB cable
- Standard LEDs + current-limiting resistors (220–330 Ω)
- Push buttons (wired straight to GND via internal pull-up)
- Rotary potentiometer (10 kΩ)

5.2 Specialized Project Modules (Introduced in Ex 3)

- **LDR / BH1750:** Ambient light sensing
- **DHT22 / BME280:** Temperature & humidity monitoring
- **PIR Sensor:** Human presence detection
- **RGB NeoPixel Ring:** Chromatic status signaling
- **Servo Motor & Buzzer:** Physical actuation & acoustic alerts

6. Course Structure Overview

Phase	Focus	Core Question	Tools
Phase A (Ex 0–3)	**The Smart Object**	What can an object sense, decide and do alone?	Wokwi, IDE
Phase B (Ex 4–6)	**The Connected Object**	What changes when an object is on the network?	Arduino Cloud
Phase C (Ex 7–10)	**The Ecosystem**	What emerges when objects cooperate?	Shared Cloud
Phase D (Ex 11–14)	**Data-Driven Design**	How can field data improve product and service?	Cloud, Sheets
Final Project (Ex 15)	**Synthesis**	Multi-object connected product-service ecosystem	All

Template for Every Exercise:

Goal → Concepts → Activities → **Design Reflection** → **Validation**

Phase A: The Smart Object

Exercises 0 – 3

Exploring local sensing, actuation, state logic, and the limits of isolation

Ex 0 — Hello Light: GPIO Output & Visual Feedback

- **Goal:** Familiarize with tools and control physical output through an LED.
- **Key Concepts:** Wokwi vs. physical board, Arduino IDE 2, `setup()`/`loop()`, `pinMode()`, `digitalWrite()`, `delay()`.
- **Activities:**
  - Run Blink in Wokwi, change rhythm, add a second alternating LED.
  - Upload sketch to physical ESP32 board.
  - Program distinct product "moods" (calm pulse, alarm, heartbeat).
- **Design Reflection:** Which blinking patterns are unambiguous to a user without instructions? What differs between simulated and real physical light?
- **Validation:** Test the three mood rhythms with a peer without advance explanation. Can the observer immediately identify which is "calm", "alarm", or "heartbeat"?

Ex 1 — The Tactile Switch: GPIO Input & Interaction

- **Goal:** Read user intention via a push button and build a responsive control loop.
- **Key Concepts:** `digitalRead()`, internal pull-up (`INPUT_PULLUP`), Serial Monitor, conditional logic (`if/else`), momentary vs. latching/toggle state, switch bouncing.
- **Activities:**
  - Wire button to GND (no external resistor required). Inspect pin state.
  - Implement direct momentary feedback (push-to-light).
  - Implement state memory: toggle lamp state ON ↔ OFF per press.
  - Observe physical contact bouncing on hardware; add debouncing.
- **Design Reflection:** Momentary (doorbell) vs. toggle (table lamp) affordances. How critical is instantaneous visual feedback for user confidence?
- **Validation:** Confirm toggle reliability across 20 rapid presses with zero bounce glitches; validate that the physical tactile sensation matches user expectations.

Ex 2 — The Continuous Dial: Analog Read & PWM

- **Goal:** Capture continuous gestures and produce nuanced modulated output.
- **Key Concepts:** ADC (12-bit, 0–4095), potentiometer voltage divider, `analogRead()`, Serial Plotter, PWM (`analogWrite()`), linear scaling with `map()`.
- **Activities:**
  - Wire potentiometer, read ADC values, inspect smooth curves on Serial Plotter.
  - Dim an LED via PWM duty cycle modulation.
  - Map ADC input (0–4095) to PWM brightness (0–255).
  - Alternative modulation: map knob position to blink tempo (50 ms–1500 ms).
- **Design Reflection:** When is a continuous dial superior to a discrete button? How do friction, throw, and knob resistance communicate product quality?
- **Validation:** Check Serial Plotter for full-range ADC linearity without dead zones; validate with a user if they can dial in a target light level by physical feel alone.

Ex 3 — The Isolated Smart Object: State & Non-blocking

- **Goal:** Synthesize primitives into a standalone prototype with distinct modes.
- **Key Concepts:** Sense → Decide → Act loop, Finite-State Machines (`OFF`, `FOCUS`, `ALERT`), non-blocking timing with `millis()`, modular project hardware.
- **Activities:**
  - Eliminate `delay()`: replace with non-blocking timestamp arithmetic.
  - Implement a 3-state machine for the studio object (e.g. Focus Lamp).
  - Integrate chosen Project Module (LDR, PIR, NeoPixel, Servo, or DHT22).
- **Design Reflection:** Compile the **Isolation Limitations List**: everything the object *cannot* do alone (know room occupancy, alert remotely, report wear).
- **Validation:** Verify zero input lag during continuous LED fading; validate that state transitions are deterministic and cycle through a studio role-play scenario.

Phase B: The Connected Object

Exercises 4 – 6

Putting the object online, binding to cloud variables, and designing remote services

The Mental Model of Arduino Cloud

> **"A cloud variable is a variable that exists in two places at the same time — on the board and in the cloud — and Arduino Cloud keeps them in sync."**

- **Device:** The physical ESP32 registered with Device ID & Secret Key.
- **Thing:** The digital twin containing cloud variables and network credentials.
- **Cloud Variables:** Defined by name, type, update policy, and permission (*Read only* for telemetry; *Read & Write* for remote control).
- **Auto-generated Skeleton:** Handles Wi-Fi connection and sync in `ArduinoCloud.update()`.

Ex 4 — Going Online: Device, Thing & First Dashboard

- **Goal:** Connect the physical prototype to the cloud and stream live sensor data.
- **Key Concepts:** Provisioning third-party devices, cloud variables (*Read only*), `thingProperties.h`, `ArduinoCloud.update()`, live dashboard widgets.
- **Activities:**
  - Register ESP32 in Arduino Cloud; configure Thing and Wi-Fi credentials.
  - Add sensor variables (e.g. `light`, `temperature`, `status`).
  - Merge Exercise 3 logic into the cloud sketch and upload.
  - Build a live monitoring dashboard with value, gauge, and status widgets.
- **Design Reflection:** Your data is now visible to the internet. What are the privacy implications? What should stay local at the edge vs. sent to the cloud?
- **Validation:** Measure latency between physical sensor changes and dashboard updates; conduct a peer privacy audit on data visible on screen.

Ex 5 — Remote Control: Mobile App & Consistency

- **Goal:** Enable remote actuation while keeping physical and digital controls consistent.
- **Key Concepts:** *Read & Write* variables, `onXxxChange()` callbacks, control widgets (switches, sliders, color pickers), **Arduino IoT Remote** mobile app.
- **Activities:**
  - Add controllable variables (e.g. `brightness`, `mode`, `lampColor`).
  - Implement callbacks to actuate LEDs/servos when cloud values change.
  - Synchronize physical inputs: pressing the hardware button updates the cloud.
  - Optimize the dashboard layout for mobile smartphones.
- **Design Reflection:** The object has two simultaneous interfaces (physical & digital). Who "wins" when both are operated at once? How do you prevent confusion?
- **Validation:** Concurrently actuate the physical knob and cloud slider; validate that both converge without feedback loops or desynchronization.

Ex 6 — The Object as a Service: Policies & Resilience

- **Goal:** Package the prototype into a reliable, secured, and well-designed service.
- **Key Concepts:** Update policies (*on change* with delta threshold vs. *periodic*), specialized variable types, connection callbacks, security of secret keys.
- **Activities:**
  - Tune update policies to minimize energy and cloud traffic.
  - Implement offline signaling on the board (e.g. slow pulse while connecting).
  - Redesign dashboards specifically tailored to distinct user personas.
  - Compile the Cloud sketch locally in Arduino IDE 2 to bypass daily quotas.
- **Design Reflection:** Which part is the product: the physical object or the digital service? Sketch the end-to-end service blueprint.
- **Validation:** Test dashboard usability with a target persona (under 30s comprehension); disconnect Wi-Fi to validate graceful offline behavior.

Phase C: The Ecosystem

Exercises 7 – 10

Loose coupling, variable synchronization, orchestration, and emergent behavior

The Shared Variable Contract

For objects built by different teams to interoperate, they must share a common schema:

Variable	Type	Permission	Purpose	Producer
`studioScene`	`int` (0–3)	Read & Write	Global room mode (Normal, Focus, Pres., Closed)	Hub, Scene Switch
`someoneAtDoor`	`bool`	Read only	Person arrived at studio entrance	Door Badge
`roomOccupied`	`bool`	Read only	Human presence in room	Presence Totem
`co2` / `noiseLevel`	`int`	Read only	Environmental air/sound quality	Studio Monitors
`<obj>Heartbeat`	`int`	Read only	Incremental liveness counter (1/min)	Every Object

Design Insight: Designing the variable contract is as fundamental as designing the physical industrial casing!

Ex 7 — Objects Talking to Objects: Variable Sync

- **Goal:** Enable direct multi-device cooperation without centralized code.
- **Key Concepts:** Variable synchronisation between Things, contract adherence, loose coupling (producer does not know consumer), events vs. commands.
- **Activities:**
  - Pair up with another group's Thing (e.g. Door Badge triggers Desk Lamp greeting).
  - Link variables using *Sync with other Things*. Write reactive callbacks.
  - Swap partners: connect to a different group's object without editing firmware.
- **Design Reflection:** What made connecting to another group's object easy or painful? How does a contract mirror physical connector standards?
- **Validation:** Validate interoperability by swapping partner objects; verify that data binding works seamlessly without code recompilation.

Ex 8 — Who is There? Heartbeats & Resilience

- **Goal:** Make the ecosystem resilient to missing, disconnected, or unpowered devices.
- **Key Concepts:** Device online/offline events, heartbeat counters, fallback strategies, **graceful degradation**, studio map overview dashboard.
- **Activities:**
  - Publish `<object>Heartbeat` and capability strings every minute.
  - Monitor partner heartbeats; switch to local fallback if silent for 3 minutes.
  - Unplug devices at random to test network fault tolerance.
  - Build a shared **Studio Map Dashboard** with live floor plan status.
- **Design Reflection:** In IoT, failures are normal daily events. How should an object communicate to users that its partner is missing without appearing "broken"?
- **Validation:** Unplug partner objects and verify that local fallback takes over within the timeout; check that the studio map dashboard clearly reflects liveness.

Ex 9 — The Orchestrator: Scenes, Hub & Services

- **Goal:** Introduce central coordination and external data feeds.
- **Key Concepts:** Coordinated scenes (`studioScene`), Hub Thing (context rules), time-based schedulers, cloud triggers, webhooks & weather services.
- **Activities:**
  - Build a studio scene selector (e.g. "Presentation": dim lamps, silence totems).
  - Create hub rules combining inputs (e.g. Empty room + window open $\rightarrow$ alert).
  - Add scheduled automation (e.g. evening studio shutdown).
  - Ingest external data: sync with a Weather Thing (rain forecast $\rightarrow$ umbrella cue).
- **Design Reflection:** Edge vs. Cloud: which rules belong in the device (fast, private, offline) vs. in the cloud (global view, multi-source context)?
- **Validation:** Trigger multi-device scenes and inspect synchronization; disconnect external internet to observe which local interactions survive.

Ex 10 — Ecosystem Design Sprint: Emergence & Conflicts

- **Goal:** Conceive, model, and resolve system-wide ecosystem interactions.
- **Key Concepts:** System mapping, emergent properties, contradictory rule conflicts, user override hierarchies, space tagging in Arduino Cloud.
- **Activities:**
  - Map all objects, variables, spatial tags, and actors in the room.
  - Identify and implement an **emergent feature** (e.g. unified comfort index).
  - Simulate and resolve a **conflict** (e.g. auto-off rule vs. reading in a corner).
  - Tag all Things by zone/role (`area:desk`, `role:sensor`).
- **Design Reflection:** Compare current system capabilities with the limitation list from Exercise 3. What new risks (lock-in, complexity) were created?
- **Validation:** Stage opposing rules simultaneously to validate override priorities; test whether the emergent comfort index aligns with actual human perception.

Phase D: Data-Driven Design

Exercises 11 – 14

Gathering field telemetry, understanding human use, predictive maintenance, and OTA evolution

Ex 11 — Remembering: Storing & Visualising Data

- **Goal:** Transition from ephemeral live streams to historical design telemetry.
- **Key Concepts:** Cloud time-series, telemetry (continuous) vs. events (discrete actions), event counters, **instrumentation plans**, CSV export.
- **Activities:**
  - Draft an instrumentation plan: connect design questions to specific metrics.
  - Instrument the sketch with event variables (e.g. `dimmerUses`, `lastMode`).
  - Build a dedicated historical "Usage Dashboard" with time-window charts.
  - Download raw CSV logs; create initial pivot tables in a spreadsheet.
- **Design Reflection:** Data only answers questions you anticipated. What would you like to know that cannot be sensed? What data must be minimized (GDPR)?
- **Validation:** Compare 24-hour logs against known physical actions to confirm no events were lost; audit metrics against data minimization standards.

Ex 12 — How is it Really Used? Field Deployment

- **Goal:** Deploy in real contexts to extract qualitative & quantitative design insights.
- **Key Concepts:** In-the-wild deployment, usage analytics, identifying "struggle" patterns, comparing intended vs. actual user behavior.
- **Activities:**
  - Deploy objects in the studio, department, or home for **at least 1 week**.
  - Analyze CSV logs for non-used features, struggle patterns, or flawed defaults.
  - Conduct short follow-up user interviews to contextualize the numbers.
  - Formulate 3 actionable redesign proposals supported by data.
- **Design Reflection:** Where did data surprise you? Where would data alone have misled you without interviewing real users?
- **Validation:** Cross-validate identified anomalies (e.g. repeated button presses) with qualitative interview quotes to ensure true user intent is understood.

Ex 13 — Keeping it Alive: Health & Maintenance

- **Goal:** Instrument products for diagnostics, reliability, and maintenance.
- **Key Concepts:** Health metrics (uptime, Wi-Fi RSSI, heap memory, battery voltage), wear counters (cycles, on-hours), drift detection, **predictive maintenance**.
- **Activities:**
  - Add persistent wear counters saved across reboots.
  - Configure multi-tier alarms (critical, major, minor, warning).
  - Build a "Fleet Health" dashboard for the space manager/technician.
  - Implement automated push/email triggers for diagnostic alerts.
- **Design Reflection:** How does maintenance telemetry alter physical industrial design (battery access, antenna placement, modular replacement parts)?
- **Validation:** Simulate hardware faults (voltage drop, signal shielding, frozen sensor); confirm alerts fire at correct severity without notification fatigue.

Ex 14 — Closing the Loop: Settings & OTA Updates

- **Goal:** Update and improve deployed products remotely based on data insights.
- **Key Concepts:** Cloud configuration variables (zero-recompile parameter tuning), A/B testing in hardware, Over-The-Air (OTA) firmware deployment.
- **Activities:**
  - Expose operational parameters as cloud sliders (e.g. `autoOffDelay`, `sensitivity`).
  - Deploy an A/B test across studio objects to evaluate two feedback variants.
  - Measure variant performance on an A/B analytics dashboard.
  - Deploy winning firmware over-the-air across the fleet.
- **Design Reflection:** A product is no longer static when sold. What does continuous evolution mean for brand trust, user agency, and ethics?
- **Validation:** Alter remote settings and confirm instant live adaptation; push an OTA update and verify smooth installation with zero manual intervention.

Final Synthesis

Exercise 15 & Cross-Cutting Themes

Ex 15 — The Final Capstone Project

- **Brief:** Design a complete **Product-Service Ecosystem** addressing a chosen context (workplace, healthcare, retail, domestic, or cultural spaces):
  1. ≥ 3 **Interconnected Objects:** Sharing a documented variable contract.
  2. **Emergent Value:** Delivering functionality impossible for a single device.
  3. **Instrumentation & Field Study:** Quantitative telemetry + user interviews.
  4. **Data-Driven Iteration:** Demonstrating at least one validated redesign (OTA / settings) justified by data.
- **Validation Requirements:**
  - Live ecosystem demonstration under real environmental conditions.
  - Interoperability test verifying error-free contract compliance.
  - Empirical proof that field data justified the redesign iteration.
  - Critical evaluation of privacy, ethical constraints, and offline degradation.

Cross-Cutting Design Themes

Throughout all phases, keep these strategic questions at the center of the design critique:

- **Privacy & Ethics:** Transparency to bystanders; data minimization by default.
- **Security:** Guarding secret keys, Wi-Fi credentials, and dashboard access.
- **Sustainability:** Power optimization (sleep modes), longevity, and circular repair.
- **Platform Dependency:** Avoiding catastrophic lock-in; defining an exit plan.
- **Interoperability:** From the studio contract to emerging standards (Matter, Home Assistant).
- **Edge vs. Cloud:** Preserving core product utility when connectivity drops.

Design with Data. Build in Ecosystems.

Ready to prototype in Wokwi, IDE 2, and Arduino Cloud.

Read Full Course Document (intro.md)

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